

A PROJECT TITLE

ON

Product Demand Forecasting Using LSTM and Gradient Boosting

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF Artificial Intelligence and Data Science.

Name of Title: Product Demand Forecasting Using LSTM and Gradient Boosting

1. Abstract:

Accurate product demand forecasting is essential for effective inventory management, supply chain optimisation, and business decision-making. Traditional forecasting methods often struggle to capture complex patterns in sales data, especially when demand is influenced by seasonal trends, promotions, customer behaviour, and market fluctuations. This project presents a hybrid demand forecasting model that combines Long Short-Term Memory (LSTM) networks and Gradient Boosting techniques to improve prediction accuracy. LSTM, a specialised type of recurrent neural network, is highly effective in learning long-term temporal dependencies from historical sales data, making it suitable for time-series forecasting. Gradient Boosting complements the LSTM model by capturing non-linear relationships between demand and additional influencing factors such as product category, pricing, promotional activities, holidays, and external variables.

The proposed approach involves collecting and preprocessing historical sales data, handling missing values, normalising features, and engineering relevant input variables. The LSTM model is trained to identify sequential patterns in demand over time, while the Gradient Boosting model learns feature-based relationships that contribute to demand variation. The predictions generated by both models are integrated to produce a more robust and reliable demand forecast. Model performance is evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

The hybrid forecasting framework aims to reduce forecasting errors, minimise stock shortages and overstock situations, and enhance inventory planning. By providing accurate demand estimates, businesses can optimise procurement, improve warehouse utilisation, reduce operational costs, and increase customer satisfaction. The proposed system is scalable and can be applied across various retail and e-commerce environments where demand patterns are dynamic and data-driven decision-making is critical. Overall, this project demonstrates how integrating deep learning and machine learning techniques can significantly enhance product demand forecasting, enabling organisations to achieve greater efficiency, profitability, and resilience in increasingly competitive markets.

2. INTRODUCTION:

Product demand forecasting is an important process that helps businesses predict future customer demand for products. Accurate forecasting enables organisations to maintain the right inventory levels, reduce costs, avoid stock shortages, and improve customer satisfaction. With the growth of retail and e-commerce, businesses require intelligent forecasting systems to make better decisions and improve supply chain efficiency.

Traditional forecasting methods often struggle to handle complex sales patterns influenced by factors such as seasonal trends, promotions, pricing, and customer behaviour. Machine learning and deep learning techniques provide more accurate predictions by learning patterns from historical sales data. Long Short-Term Memory (LSTM) is effective for analysing time-series data, while Gradient Boosting captures nonlinear relationships between different demand-related features.

This project proposes a hybrid forecasting model that combines LSTM and Gradient Boosting to improve prediction accuracy. The system processes historical sales data, trains both models, and generates reliable demand forecasts. The proposed approach helps businesses optimise inventory management, reduce operational costs, and support data-driven decision-making, making it suitable for retail, e-commerce, and manufacturing industries.

3. SOFTWARE REQUIREMENTS:

- 1) Operating System: Windows 10/11
- 2) Programming Language: Python 3.x
- 3) IDE: Jupyter Notebook / VS Code / Google Colab
- 4) Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- 5) Dataset: CSV File (.csv)



Signature of the Guide



Course Instructor